

PAPER NAME

I2ITCON26_5351.pdf

AUTHOR

-

WORD COUNT

4720 Words

CHARACTER COUNT

28601 Characters

PAGE COUNT

8 Pages

FILE SIZE

361.0KB

SUBMISSION DATE

Apr 21, 2026 12:30 PM GMT+5:30

REPORT DATE

Apr 21, 2026 12:30 PM GMT+5:30**● 23% Overall Similarity**

The combined total of all matches, including overlapping sources, for each database.

- 16% Internet database
- 21% Publications database
- Crossref database
- Crossref Posted Content database
- 0% Submitted Works database

Dynamic Design Education – Daily Dose of Engineering

Prof. Deepika Walanjkar

Department of Information Technology
International Institute of Information
Technology
Pune, India
deepikaw@isquareit.edu.in

Atharva Hegade

Department of Information Technology
International Institute of Information
Technology
Pune, India
atharvahegade007@gmail.com

Swapnil Fuse

Department of Information Technology
International Institute of Information
Technology
Pune, India
swapnilfuse85@gmail.com

Raghav Daga

Department of Information Technology
International Institute of Information
Technology
Pune, India
raghavdaga1104@gmail.com

Sanket Dongre

Department of Information Technology
International Institute of Information
Technology
Pune, India
sanketdongre33@gmail.com

1. Abstract—

The Dynamic Design Education (DDE) project is a strategic initiative aimed at creating a comprehensive and dynamic web application for engineering students. The core mission is to bridge the significant gap between theoretical academic knowledge and the practical demands of the modern technology industry. By targeting this critical audience, the DDE seeks to foster technical proficiency, enhance interview readiness, and develop problem-solving capabilities that are directly relevant to professional engineering roles. The platform's unique value proposition is its holistic and integrated ecosystem. Unlike existing solutions that address only a single aspect of the learning journey, such as course platforms (e.g., Coursera), coding practice sites (e.g., LeetCode), or community forums (e.g., Reddit), the DDE is designed as a single, "one-stop-shop" solution. It integrates daily, bite-sized microlearning content, adaptive learning paths powered by sophisticated algorithms, advanced career-readiness tools such as an ATS resume checker, and a vibrant, interactive community platform. This comprehensive approach streamlines the user's educational and career development journey.

2. Introduction

2.1 Background and Motivation

The modern learning landscape is defined by a knowledge explosion and rapid digitization, leading to exponential availability of open educational resources (OERs). Particularly since the outbreak of COVID-19, there has been a heightened demand for robust alternative methods to deliver educational materials remotely, resulting in a swift migration to online platforms that combine learning management systems and OER delivery across multiple channels, such as YouTube. However, traditional educational models, which often provide a "one-size-fits-all" approach, frequently lead to a suboptimal learning experience that fails to address individual proficiency, learning pace, or prior knowledge. The

fundamental motivation of this project is to transform learning platforms from mere content delivery systems into powerful tools for developing lasting knowledge mastery, critical thinking, and problem-solving skills. The implementation of truly effective personalized learning faces significant technical challenges. A central issue is the cold-start problem, which occurs when the system lacks sufficient historical data to provide accurate and relevant recommendations for new users or recently added content. This challenge is particularly acute in micro-open learning environments, where learners engage in fragmented time spans, and the system knowledge of fresh learners is initially sparse. The scarcity of initial data makes it infeasible to construct comprehensive learner models instantaneously, which is required for effective offline computation systems such as Microlearning as a Service (MLaaS). Without an adequate solution, this data sparsity can result in irrelevant recommendations, reducing user satisfaction and potentially causing learners to abandon the platform entirely. To overcome these limitations, we propose an innovative approach centered on online computation and a hybrid algorithmic architecture tailored for fine-grained personalization and memory reinforcement. The core project (e.g., the DDE platform or MLaaS approach) aims to deliver adaptive learning paths and assessments that bridge the gap between theoretical knowledge and practical industry demands for technical freshers in the field. The proliferation of Open Educational Resources and the global trend toward remote delivery, accelerated by events such as COVID-19, have highlighted the urgent need for robust alternative educational delivery methods.

2.2 The Challenge of Adaptive Learning and Cold Start

Implementing truly effective personalized learning faces significant technical hurdles. A central issue is the cold start problem, which occurs when the system lacks sufficient historical data to provide accurate and relevant recommendations for new users or recently added content. This challenge is particularly acute in micro-open learning environments, where learners engage in fragmented time

spans and system knowledge of fresh learners is initially sparse. The scarcity of initial data makes it infeasible to construct comprehensive learner models instantaneously, which is required for effective offline computation systems like Microlearning as a Service (MLaaS). Without an adequate solution, this data sparsity can result in irrelevant recommendations, reducing user satisfaction and potentially causing learners to abandon the platform.

2.3 Project Goal and Novel Approach

To overcome these limitations, we propose an innovative approach centred on online computation and a hybrid algorithmic architecture tailored for fine-grained personalization and memory reinforcement. The core project (e.g., the DDE platform or MLaaS approach) aims to deliver adaptive learning paths and assessments that bridge the gap between theoretical knowledge and practical industry demands for technical freshers.

3.Related Work

3.1 Adaptive Learning and Personalized Systems

The proliferation of Open Educational Resources and the global trend toward remote delivery, accelerated by events such as COVID-19, have highlighted the urgent need for robust alternative educational delivery methods. Traditional, static instruction models often fail to adapt to individual proficiency, prior knowledge, or fragmented time spans characteristic of modern learning environments.

Adaptive Learning Components: Research into building effective adaptive systems focuses heavily on modelling both the user and the content:

A. Learner Profiling: Effective adaptation relies on accurate learner models. Prior research has established comprehensive learner models, sometimes annotated using semantic approaches, incorporating intellectual, non-intellectual, and external contextual factors relevant to micro-open learning experiences and outcomes. Other works focus on predicting learning styles and automatic learner classification. However, deploying such comprehensive models is challenging in new environments due to initial data scarcity. This challenge motivates the use of simplified, lightweight learner profiles for swift decision-making during the platform's initialization phase.

B. Content Representation: Content organization, specifically for OERs, benefits significantly from semantic Web technologies and ontological modelling. To support fine-grained adaptation, OER ontologies can be augmented to represent micro-OERs as self-describing nodes annotated with metadata, including parameters like the didactic model, interaction type, and functional attributes such as Quality of Service (QoS).

3.2 Recommendation Systems and the Cold Start Problem

The development of personalized microlearning pathways is constrained by the cold start problem—the lack of historical data for new users or continuously released new content (OERs). This data sparsity restricts the performance of data-hungry offline computation systems, such as the proposed Microlearning as a Service (MLaaS) framework, by making instantaneous and accurate learner model construction infeasible.

Past attempts to mitigate the cold start issue include focusing on user profiling, often leveraging established datasets like MovieLens. However, researchers have noted

that migrating recommenders trained on entertainment data sets (like movies) do not show convincing performance on educational data, largely due to the inherent isolation of the educational domain and the scarcity of suitable large-scale data assets. This discrepancy leads to a focus on solutions requiring less explicit user rating information, which is often scarcely known at the beginning of a learning activity.

To overcome these challenges, existing research proposes applying online computation coupled with heuristic solutions during the system's warm-up stage. Heuristic algorithms utilize lightweight profiles and domain-specific rules to generate relevant initial recommendation lists and integrate newly released micro-OERs into established learning paths, thereby improving their discoverability.

3.3 Memory Reinforcement and Adaptive Assessment

Effective adaptive systems must move beyond content delivery to focus on long-term knowledge retention, primarily supported by cognitive science principles:

A. Spaced Repetition: The concept of scheduling repetition at progressively increasing intervals is rooted in the work of Ebbinghaus, who hypothesized that the human forgetting rate increases exponentially with time. The goal is to review items precisely as they are about to be forgotten, ensuring maximum retention. Foundational algorithms include the Pimsleur Method (fixed schedule) and the Leitner System (more adaptive spacing based on performance). More recently, Half Life Regression (HLR) has been proposed as a successful generalization of these models for language learning platforms.

B. Adaptive Algorithms for Retention: To address the varying needs of diverse learners, models like the Spaced Repetition for Slow Learners (SRSL) algorithm have been developed. SRSL computes a learner's score based on fine-grained factors such as response time, question difficulty, and prerequisite dependency. It integrates prerequisite relationships via a directed acyclic graph, allowing positive reinforcement (rewarding prerequisite questions when a subsequent question is answered correctly) and penalty application for closely associated topics when mistakes occur.

C Adaptive Quizzing: In adaptive testing, two models are frequently combined for robustness: Bayesian Knowledge Tracing (BKT), which probabilistically tracks a learner's mastery of specific skills over time, and Item Response Theory (IRT), which estimates the intrinsic characteristics of test questions, such as difficulty and discrimination. Combining these models is critical, as relying solely on IRT can present a significant "cold-start" problem due to the need for large amounts of response data for accurate item calibration.

3.4 Algorithmic Intelligence for Path Planning

The generation of dynamic, optimal learning sequences aligns with complex decision-making problems traditionally addressed by game Artificial Intelligence (AI) and planning algorithms.

3.4.1 Monte Carlo Algorithm

The Monte Carlo Tree Search (MCTS) algorithm is a powerful decision-making approach used primarily for designing artificial intelligence (AI) agents in games and solving complex, sequential decision problems. The method fundamentally relies on an intelligent tree search

mechanism that effectively balances the need for exploration of novel, untested actions with the exploitation of actions that have yielded the best results so far. Rooted in the techniques of random sampling, MCTS conducts simulations, often called playouts, and systematically stores performance statistics for various actions, allowing it to make increasingly educated choices across iterations. Historically, MCTS was first introduced for creating computer players in the game of Go and is inherently a heuristic method, only requiring knowledge of the problem's rules, although domain-specific heuristics can be integrated to improve efficiency and convergence.

The operation of the classical MCTS algorithm proceeds through four distinct, sequential phases in each iteration:

A. Selection: The algorithm navigates the portion of the search tree already built and stored in memory, starting from the root node. At each step, the next node is chosen based on a selection policy (or tree policy) until a leaf node or a terminal state of the problem is reached.

B. Expansion: If the end point of the selection phase is a non-terminal leaf node, the algorithm enters the expansion phase, where it adds at least one new child node to the tree, corresponding to the state reached by the last performed action.

C. Simulation (Playout): Starting from the newly added node, a complete simulation of the problem is executed, often using a default random policy, until a terminal state is reached and the final score or payoff is determined.

D. Backpropagation: The score obtained from the simulation is then propagated back up the path from the terminal state to the root. This process updates the stored statistics, such as visit counts and average rewards, for all nodes and modelled agents along the traversal path.

The mechanism for balancing exploration versus exploitation, which is critical to MCTS's success, is most commonly implemented using the Upper Confidence Bounds applied for Trees (UCT) formula, leading the approach to often be referred to as MCTS/UCT. UCT enables an asymmetric search of the game tree, focusing computational resources along lines of play deemed most promising. The selection of the optimal action is determined by a formula that incorporates (the average result of playing action in state) and an exploration term that involves the logarithm of the state visit count divided by the action visit count, regulated by an exploration constant. Beyond UCT, other selection policies exist, such as Thomson sampling and the Exploration-Exploitation with Exponential weights (EXP3). Furthermore, enhancements like Rapid Value Action Estimation (RAVE) have been developed to mitigate the "cold start" effect by maintaining an auxiliary score () that is updated for all actions taken during the simulation phase, effectively sharing action values across related subtrees. Applications of MCTS extend widely beyond the realm of games into complex planning and decision-making systems.

• Automated Planning: MCTS is frequently applied in automated planning problems, which are typically formulated as a Markov Decision Process (MDP) or a Partially Observable MDP (POMDP). Specific industrial and academic challenges where MCTS has been utilized include scheduling problems (such as forest harvest scheduling) and various formulations of Vehicular Routing

Problem (VRP).

Chemical Synthesis: MCTS has achieved novel results when combined with deep neural networks for computer-aided retrosynthesis planning (planning chemical syntheses), in an approach termed 3N-MCTS.

Adaptive Learning Paths: In modern personalized learning platforms, MCTS is proposed for dynamically constructing optimal learning sequences for users. This process frames learning as a search problem on a knowledge graph where learning units are represented as nodes and prerequisites as edges. By intelligently managing the balance between exploring new content and exploiting (reinforcing) mastered topics, MCTS moves beyond static curricula to truly personalize the learning journey.

The MCTS algorithm offers several key advantages for inclusion in a research paper, but also presents important challenges. Its strengths include its ability to build the search tree asymmetrically, concentrating resources efficiently. It is also known as an anytime algorithm, meaning execution can be stopped at any moment to return the best solution found thus far. Moreover, the method is inherently easy to parallelize; common parallelization strategies include Leaf Parallelization, Root Parallelization, and Tree Parallelization. When integrated with Machine Learning (ML), MCTS achieves superior results (as seen in Go) and benefits from the approximation and generalization capabilities of ML models, which can evaluate states never encountered during previous simulations. Conversely, a primary challenge is that MCTS is a search-based algorithm and is prone to search space explosion. This issue is often addressed through methodological modifications, such as simplifying the problem model or reducing the action space. Another strategy to mitigate high computational cost is to run MCTS offline for a sufficient duration to generate high-quality training data, which is then used to train a computationally lighter ML model for online use.

3.4.2 MCTS in Personalized Learning (DDE Platform)

In the context of personalized education, MCTS is used to generate adaptive learning paths.

Role in Learning Engine: The DDE Learning Engine leverages MCTS to dynamically construct optimal learning sequences for users.

Framing the Problem: The learning process is explicitly framed as a search problem on a knowledge graph.

Graph Components: In this graph, learning units are represented as nodes, and prerequisites are represented as edges.

Balancing Learning: MCTS intelligently explores this graph, ensuring a balance between "exploration" (introducing new, less-tested topics) and "exploitation" (reinforcing mastered topics) to create a personalized learning journey.

Differentiation: This novel application of a game-playing algorithm is a significant differentiator for the platform, allowing the system to move beyond static curricula and truly personalize the experience.

Modifications and Challenges

The vanilla MCTS algorithm often fails to deliver expected performance in complex settings with other techniques.

Hybridisation: MCTS is frequently hybridised with other techniques, such as Machine Learning (ML) and

Evolutionary Methods. For instance, ML models (like neural networks) can be used to model the value function (approximating the game outcome) and the policy function (informing which action to choose).

Performance Improvements: Methods exist to enhance the base algorithm, such as Rapid Value Action Estimation (RAVE), which minimizes the "cold start" effect by using action values gathered from all actions during simulation, not just selected ones.

Addressing Search Space Explosion: MCTS is prone to search space explosion (combinatorial complexity). Modifications often focus on Action Reduction (eliminating poor actions to prevent the tree from growing sideways) or simplifying the problem model.

Parallelization: MCTS is inherently easy to parallelize, which is a common strategy to increase the overall number of iterations performed within a computational budget. The three major approaches to parallel MCTS are Leaf Parallelization, Root Parallelization, and Tree Parallelization.

3.4.3. Bayesian Knowledge Tracing (BKT) and Item Response Theory (IRT)

The integration of Bayesian Knowledge Tracing (BKT) and Item Response Theory (IRT) models forms a robust hybrid approach specifically designed for adaptive quizzing and personalized assessment. This synergistic strategy addresses the limitations of each model when used in isolation, leading to a more accurate and scalable assessment engine, particularly suitable for platforms like the Daily Dose of Engineering (DDE).

Below is a detailed breakdown of this integration, focusing on the distinct roles, mechanisms, and benefits of each model:

A. Bayesian Knowledge Tracing (BKT)

BKT is the core model for tracking a learner's skill mastery over time.

Function: BKT is a probabilistic model that is specialized as a Hidden Markov Model. It infers and tracks whether a learner has mastered a specific skill, representing the learner's knowledge state as a binary variable (they either know the skill or they do not).

Skill Tracking: The system continuously updates its belief (the mastery probability, p) regarding a user's skill mastery after every question is answered, regardless of whether the answer was correct or incorrect.

Key Parameters: BKT relies on four skill-specific parameters that must be calibrated for each knowledge component:

(p -init): The initial probability that a student knows the skill. (p -transit): The probability that a student learns a skill after an opportunity to apply it. (p -slip): The probability that a student makes an error even when they genuinely know the skill (accounting for human error). (p -guess): The probability that a student correctly answers a question without actually knowing the underlying skill.

Limitation in Isolation

Standard BKT models typically do not account for the difficulty of the questions presented. A correct response to a simple question is treated the same as a correct response

to a difficult one in terms of updating the mastery probability, which can restrict the model's accuracy.

B. Item Response Theory (IRT)

IRT focuses on the inherent characteristics of the questions (items) themselves, providing context for the difficulty of the assessment.

Function: IRT models the probability of a test-taker answering a question correctly based on the item's characteristics. It complements BKT by allowing the system to select the "most informative" question to present next.

Key Parameters: IRT models rely on item-specific parameters, commonly including:

Difficulty (b): The ability level (θ) at which a test-taker has a 50% chance of answering the item correctly.

Discrimination (a): How well an item differentiates between examinees with ability levels just below and just above the item's difficulty.

A guessing factor (c) may also be included to account for random correct guesses on multiple-choice questions.

Limitation in Isolation: The major challenge of implementing IRT, especially in new platforms, is the "cold-start" problem. IRT requires a large quantity of pre-collected data (potentially hundreds of responses per item) to accurately calibrate and refine the difficulty and discrimination parameters of each question.

C. The Hybrid BKT/IRT Integration Approach

The combined model maximizes accuracy and addresses the cold-start data challenge by ensuring a continuous feedback loop between the question quality (IRT) and the learner state (BKT).

Mechanism of Integration: The core integration involves using the IRT-derived difficulty parameter (the 'b' value) of a question to refine the BKT update process. This means that if a user correctly answers a particularly challenging question (high IRT difficulty), their estimated mastery probability (p) for the associated skill should increase more significantly than if they answer an easy question correctly.

Addressing Cold-Start: This hybrid approach is particularly effective for new platforms (like DDE) with a nascent user base:

1. Initial Setup: Initially, subject matter experts can manually assign an estimated difficulty level to each question. This preliminary difficulty data informs the BKT model from the start, mitigating the immediate data scarcity.
2. Calibration and Refinement: As the system collects more user response data, IRT is continuously used to calibrate and refine the actual difficulty and discrimination parameters of each question.
3. Enhancement: These refined IRT parameters are then used to enhance the accuracy of the BKT mastery model, making its mastery probability updates more sensitive to item difficulty.

Adaptive Assessment: The hybrid model provides a robust foundation for real-time adaptive assessment. By tracking individual skill mastery (BKT) and knowing the precise characteristics of the available questions (IRT), the system can select the optimal question to maximize the efficiency and accuracy of assessment.

This comprehensive, hybrid modelling approach ensures the system provides precise and meaningful personalization

while simultaneously gathering the necessary data for long-term algorithmic sophistication.

4.Methodology

4.1 System Architecture

The system architecture is detailed across several systems described in the sources, most notably the Dynamic Design Education (DDE) platform and the Microlearning as a Service (MLaaS) framework, as well as general e-learning cloud structural designs. Design Education (DDE) Platform Architecture Core Technology Stack Architectural Style: Microservices architecture, breaking the application into smaller, independent services.

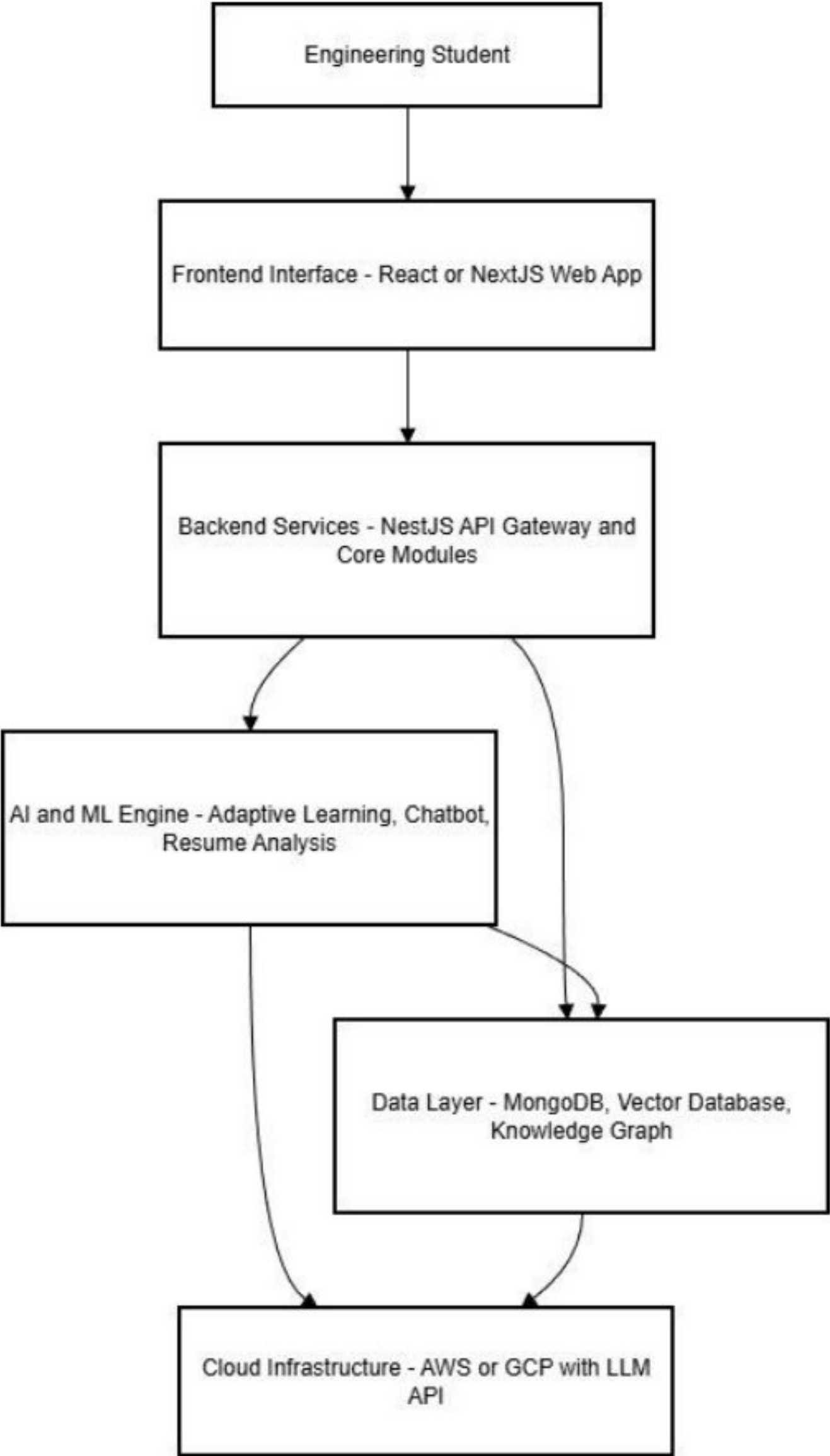


Fig. 1. Proposed System Architecture

Backend Core: NestJS, a progressive Node.js framework, is used for core business logic, providing a robust foundation and native support for the microservices architectural style. Specialized Logic: Dedicated Python microservices are employed for advanced machine learning (ML) functionalities, leveraging Python’s rich data science ecosystem.

Frontend: The user interfaces are built using React.
Data Layer: MongoDB (a NoSQL document database) is the primary data layer due to its flexible schema, suited for

Service	Responsibility	Service
User/Auth Service	Manages user profiles, authentication, and authorization.	User/Auth Service
Content Service	Handles all learning materials, including micro-notes, quizzes, and dynamic learning paths.	Content Service
Community Service	Manages the social-media-style feed and real-time interactions.	Community Service
AI/ML Service	A collection of Python microservices responsible for the AI chatbot and the ATS resume checker.	AI/ML Service

diverse data types like user profiles, content, and community posts. Prisma is used as the Object Data Model (ODM) layer for consistency and developer productivity. System Components and Data Flow All client requests interact with the system through a single API Gateway, which handles security, authentication (using JWT or OAuth 2.0), and routing to the appropriate backend service. The architecture is composed of several autonomous microservices:

Key Algorithmic Subsystems AI/ML Service The DDE learning engine integrates sophisticated algorithms, with the complex calculations routed to the Python microservice:

- Adaptive Quizzing:** Uses a hybrid Bayesian Knowledge Tracing (BKT) and Item Response Theory (IRT) model to track skill mastery and select optimal questions.
- Personalized Flashcards:** Uses an Anki-inspired adaptation of the Super Memo SM-2 algorithm for dynamically scheduling review intervals based on user performance, ensuring long-term memory reinforcement.
- Personalized Learning Paths:** Utilizes a Monte Carlo Tree Search (MCTS) algorithm operating on a foundational knowledge graph to dynamically construct optimal learning sequences.
- AI Chatbot:** Implemented using a Retrieval-Augmented Generation (RAG) architecture. This design grounds the Large Language Model (LLM) in DDE’s proprietary knowledge, which is pre-processed into vectors and stored in a vector database (or MongoDB Atlas Vector Search). Real-Time Communication. The Community Service utilizes a hybrid real-time communication strategy:

5. Server-Sent Events (SSE): Used for unidirectional news feed updates (a "server push" or "fan-out on write" model).

6. WebSockets: Employed for bidirectional, low-latency communication, suitable for features like live chat.

7. Message Broker: Services are underpinned by a message broker like Kafka for high-throughput, durable event streams.

4.2 Deployment

Implementing e-learning through cloud computing brings several challenges for educational institutions. Major issues include security concerns, limited internet bandwidth, and the difficulty of convincing decision-makers to adopt new systems. Traditional and cloud-based learning management systems also differ in areas like teaching methods, content handling, and student evaluation, requiring both educators and learners to adapt.

Studies emphasize that although cloud computing reduces hardware and software costs and offers scalability, institutions must plan transitions carefully. The COVID-19 pandemic further highlighted the importance of cloud platforms for maintaining quality education and enhancing teacher productivity through collaborative and gamified tools.

Research shows that successful adoption of cloud computing in higher education depends on factors like government support, internal readiness, IT infrastructure, and collaboration with cloud providers. Systematic reviews also identify ongoing issues such as interoperability, usability, and a lack of strong empirical studies in this field. Overall, cloud-based e-learning holds great potential to improve accessibility and collaboration in education, but more focused research and better implementation strategies are needed to ensure effective, secure, and user-friendly adoption.

The MLaaS framework is proposed as a method to realize adaptive microlearning through Open Educational Resources (OERs).

Service Model: MLaaS is a Software as a Service (SaaS) framework organized over a cloud.

Resource Integration: The system operates by having a Master Service (MLaaS) that actively searches and locates all available OERs and their containers (i.e., associated web services).

Computation Integration: MLaaS jointly uses online computation (for cold start problems) and offline computation (for executing adaptive computations using profiles) to decide learning resource delivery.

Semantic Integration: For effective integration and interoperability, MLaaS is underpinned by an augmented micro OER ontology. Utilizing a Semantic Web-based approach helps ensure that heterogeneous data from multiple OER systems can be mutually linked with better connectivity and interoperability. This ontological knowledge handles both offline and online computation, supplying preliminary knowledge to both the Adaptive Engine and the Cold Start Engine.

5 References

[1] *Dialogue Management in Conversational Systems: A Review of Approaches, Challenges, and Opportunities* Hayet Brabra, Marcos Báez, Boualem Benatallah, Senior

Member, IEEE, Walid Gaaloul, Sara Bouguelia, and Shayan Zamanirad

[2] *Adapting New Learners and New Resources to Micro Open Learning via Online Computation* Geng Sun, Member, IEEE, Wei Wei, Senior Member, IEEE, Tingru Cui, Dongming Xu, Shiping Chen, Senior Member, IEEE, Alex Shvonski, Li Li, Member, IEEE, Jun Shen, Senior Member, IEEE, and Soheila Garshasbi

[3] *E-Learning-Based Cloud Computing Environment: A Systematic Review, Challenges, and Opportunities* HANA ELJAK¹, ASHRAF OSMAN IBRAHIM^{2,3}, (Senior Member, IEEE), FAKHRELDIN SAEED⁴, IBRAHIM

[4] *Spaced Repetition for Slow Learners* Devansh P. Shah, Nikhil M. Jagtap, Shloka S. Shah and Anant V. Nimkar Department of Computer Engineering Sardar Patel Institute of Technology Mumbai, India

[5] O. K. J. Mohammad, "Recent trends of cloud computing applications and services in medical, educational, financial, library and agricultural disciplines," in *Proc. 4th Int. Conf. Frontiers Educ. Technol.*, 2018

[6] J. Riahi, "E-learning systems based on cloud computing: A review," *Proc. Comput. Sci.*, vol. 62, pp. 352–359, Jan. 2015, doi: 10.1016/j.procs.2015.08.415.

[7] S. Azouzi, S. A. Ghannouchi, and Z. Brahmi, "Study of e learning system based on cloud computing: A survey," in *Proc. Int. Conf. Intell. Syst. Design Appl. Cham, Switzerland: Springer*, 2018, pp. 532–544, doi: 10.1007/978-3-030-16660-1_52.

[8] B. S. Shukur, M. K. A. Ghani, and M. Burhasnuddin, "A cloud computing framework for higher education institutes in developing countries (CCF_HEI_DC)," in *Intelligent and Interactive Computing. Cham, Switzerland: Springer*, 2019, pp. 397–409, doi: 10.1007/978-981-13-6031-2_24.

[9] E. Leloglu, T. Ayav, and B. G. Aslan, "A review of cloud deployment models for e-learning systems," in *Proc. 43rd Annu. IEEE/IFIP Int. Conf. Dependable Syst. Netw. (DSN)*, Jun. 2013, pp. 1–2, doi: 10.1109/DSN.2013.6575331.

[10] M. A. H. Masud and X. Huang, "An e-learning system architecture based on cloud computing," *World Academy Sci., Eng., Technol.*, vol. 62, pp. 74–78, Jan. 2012.

[11] A. El Mhouti, M. Erradi, and A. Nasseh, "Using cloud computing services in e-learning process: Benefits and challenges," *Educ. Inf. Technol.*, vol. 23, no. 2, pp. 893–909, Mar. 2018, doi: 10.1007/s10639-017-9642-x.

[12] M. Ketel, "E-learning in a cloud computing environment," in *Proc. IEEE SOUTHEASTCON*, Mar. 2014, pp. 1–2, doi: 10.1109/SECON.2014.6950728.

[13] J. Hylen, D. V. Damme, F. Mulder, and S. D' Antoni, "Open educational resources: Analysis of responses to the OECD country questionnaire," *OECD, Paris, France, OECD Educ. Working Papers 76*, Jun. 2012. [14] G. Sun, T. Cui, S. Chen, W. Guo, and J. Shen, "MLaaS: A cloud system for mobile micro learning in MOOC," in *Proc. IEEE Int. Conf. Mobile Services*, New York, NY, UAS, Jun. 2015, pp. 120–127.

[15] G. Sun, T. Cui, D. Xu, H. Chen, S. Chen, and J. Shen, "Assisting open education resource providers and instructors to deal with cold start problem in adaptive micro learning: A service oriented solution," in *Proc. IEEE Int.*

Conf. Services Comput. (SCC), Honolulu, HI, USA, Jun. 2017, pp. 196–203.

[16] G. Sun, T. Cui, G. Beydoun, S. Chen, D. Xu, and J. Shen, “Organizing online computation for adaptive micro open education resource recommendation,” in *Proc. 16th Int. Conf. Web Learn. (ICWL)*, CapeTown, South Africa, Sep. 2017, pp. 177–182.

[17] A. Corbi and D. Burgos, “OERaaS: Open educational resources as a service with the help of virtual containers,” *IEEE Latin Amer. Trans.*, vol. 14, no. 6, pp. 1–7, Aug. 2016.

[18] R. Hernandez, C. Gutl, and H. R. Amado-Salvatierra, “Cloud learning activities orchestration for MOOC environments,” in *Proc. 3rd Int. Workshop Learn. Technol. Educ. Cloud*, Santiago, Chile, Sep. 2014, pp. 25–36.

● 23% Overall Similarity

Top sources found in the following databases:

- 16% Internet database
- 21% Publications database
- Crossref database
- Crossref Posted Content database
- 0% Submitted Works database

TOP SOURCES

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Geng Sun, Wei Wei, Tingru Cui, Dongming Xu, Shiping Chen, Alex Shvo... Crossref	7%
2	umpir.ump.edu.my Internet	6%
3	Maciej Świechowski, Konrad Godlewski, Bartosz Sawicki, Jacek Mańdz... Crossref	2%
4	jsju.org.cn Internet	1%
5	simplilearn.com Internet	<1%
6	Devansh P. Shah, Nikhil M. Jagtap, Shloka S. Shah, Anant V. Nimkar. "S... Crossref	<1%
7	ebin.pub Internet	<1%
8	Hayet Brabra, Marcos Baez, Boualem Benatallah, Walid Gaaloul, Sara B... Crossref	<1%

9	Lecture Notes in Computer Science, 2004.	<1%
	Crossref	
10	iucel2025.utm.edu.my	<1%
	Internet	
11	ro.uow.edu.au	<1%
	Internet	
12	api.deepai.org	<1%
	Internet	
13	archive.org	<1%
	Internet	
14	hdl.handle.net	<1%
	Internet	
15	"Digital Technologies and Applications", Springer Science and Business...	<1%
	Crossref	
16	Hana Eljak, Ashraf Osman Ibrahim, Fakhreldin Saeed, Ibrahim Abaker T...	<1%
	Crossref	
17	Morrison, Breanna. "Comparing Elo, Glicko, IRT, and Bayesian IRT Stati...	<1%
	Publication	
18	"Intelligent and Interactive Computing", Springer Science and Business...	<1%
	Crossref	
19	Guillaume M. J. -B. Chaslot. "Parallel Monte-Carlo Tree Search", Lectur...	<1%
	Crossref	
20	Tingyin Wong, Hongwei Zheng. "Adaptation Mechanisms of Hong Kon...	<1%
	Crossref posted content	

- 21

Urmila Shrawankar, Prerna Mishra. "Cloud Computing for Smart Educat... <1%
Publication

- 22

researchgate.net <1%
Internet

- 23

Joel Mandlazi, Ashwini Jadhav, Ritesh Ajoodha. "Evaluating Deep Sequ... <1%
Crossref